

AEGIS

CRYPTOGRAPHIC PROVENANCE FOR BUY CANADIAN PROCUREMENT

Origin you can **prove** — not just claim.

THE PROBLEM

“BUY CANADIAN” MATTERS — BUT NOBODY CAN VERIFY IT

1. Security you can't see

Critical electronics — chips, flight controllers, radios — sourced abroad can hide undisclosed parts or firmware-level backdoors. For defence and dual-use systems, an origin you can't verify is a risk you can't manage.

2. Sovereignty & jobs

Low-cost offshore labour and opaque tiers undercut Canadian industry. Procurement dollars meant to build domestic capability leak abroad when no one can check where value was actually added.

3. A claim, not a proof

Today “Made in Canada” rests on supplier self-reporting — easy to misrepresent, impossible to audit once a product has passed through many hands.

The Legal Bar

Product of Canada

98%

≥98% of direct costs in Canada + last substantial transformation here.

Made in Canada

51%

≥51% of direct costs in Canada + last substantial transformation here.

THE SOLUTION

AEGIS – “TRUST, BUT VERIFY” FOR ORIGIN

Don't take the supplier's word for it. Submit a product's full attestation chain; AEGIS returns its Canadian-content %, its legal designation, whether the chain is valid, and every anomaly it found — built on the organizer's **byte-exact reference crypto**.

① Verify every signature

Each contribution is an **Ed25519-signed attestation** checked against a registered supplier key, and **hash-linked** to the exact content of the inputs it consumed — a tamper-evident chain of custody.

② Compute the designation

Walk the supply graph, sum direct costs by **country of work**, and decide **Product of Canada / Made in Canada** / none against the legal thresholds — deterministically.

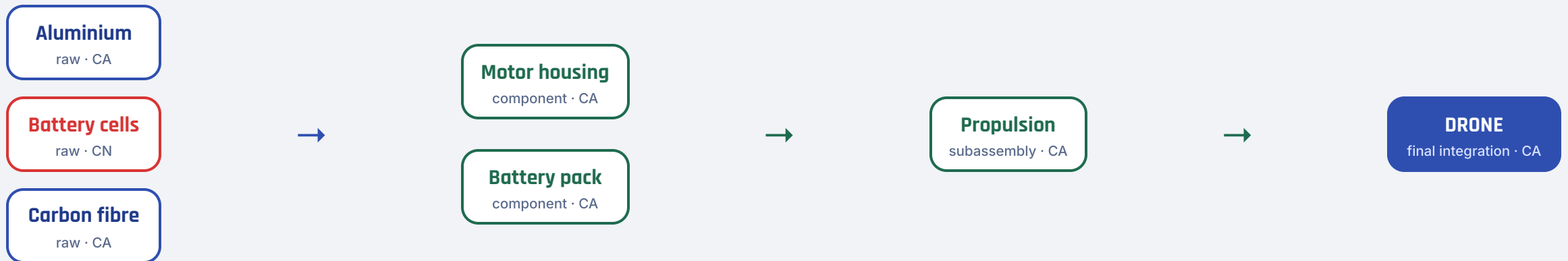
③ Detect what self-reporting hides

Catch integrity, replay, quantity, structural and statistical attacks — and stay standing on incomplete data. **Stateless, one-command Docker deploy**.

HOW IT WORKS

FROM RAW MATERIAL TO FINISHED DRONE

Each contribution is a signed attestation referencing the inputs it consumed — forming a directed graph from raw materials up to the finished product. Trace any product back to its origins.



AEGIS – SYSTEM ARCHITECTURE

TAN

Issuance → trust anchors → stateless verification engine → procurement decision

① ISSUANCE LAYER



Supplier portal

Each tier signs its contribution — action, country of work, costs, inputs consumed.



Ed25519 signing

Browser-side keypair; signs the canonical form of the attestation.



Attestation

References parents by **content hash** → hash-linked DAG across tiers.



Product chain

Leaf product + all ancestors, submitted together.

② VERIFICATION ENGINE · POST /verify (STATELESS · DOCKER)

AEGIS /verify

built on the organizer's byte-exact reference_lib

1

Build DAG

Order-independent; from parents

2

Verify sigs

Ed25519 + hash links

3

Canadian %

cost by country of work

4

Designate

98% / 51% thresholds

🚩 Anomaly detection – 5 categories

signature_invalid

signature_unknown_supplier

parent_hash_mismatch

mass_balance_violation

circular_reference

dangling_parent

timestamp_inversion

unit_mismatch

replay_within_chain

replay_cross_chain

transformation_implausible

statistical outliers (learned)

incomplete-data → graceful

→ { canadian_content_percentage, designation, chain_valid, anomalies[] }

③ TRUST ANCHORS



Supplier registry

supplier_id → public key (69 suppliers). Verified identities.



Anchor registry

Signed ledger of genuine attestations (3,147). Catches rewrite & cross-product replay.



④ CONSUMPTION



Purchaser scan

QR → digital product passport. Plain-language verdict.



Procurement

"Made in Canada · 58.4% · verified." Buy-Canadian, enforceable.

WHAT AEGIS ANSWERS

TAN

A buyer scans a product and gets a procurement-ready answer — no expertise required.

Does it qualify?

Made in Canada or **Product of Canada**, computed directly from the chain against the legal thresholds.

How Canadian is it?

The **exact percentage** of the product's direct cost performed in Canada — attributed per attestation.

Can I trust it?

Every signature verified end-to-end; **any tampering is flagged** and classified.

Example: "Made in Canada · 58.4% Canadian · chain verified · 0 anomalies"

TAMPER-EVIDENT

WHAT SELF-REPORTING CAN'T CATCH

Every supplier private key ships in the kit — so an attacker can sign a *false* chain. A valid signature isn't enough; AEGIS reasons about whether a chain is **internally consistent** and **physically plausible**, and classifies each violation.

Integrity / forgery

Modified after signing, or signed by a key not in the verified registry.

Replay & reuse

One product's attestation dropped into another's chain, or reused across products.

Quantity violations

A chain consumes more material than upstream suppliers ever produced — summed globally across diamonds.

Structural problems

Missing references, broken hash links, cycles, impossible orderings.

Statistical outliers

Legal on every rule, yet anomalous vs. how genuine chains look — learned from 1,000 chains.

Incomplete data

Missing fields flag uncertainty — the verifier still returns a useful answer without falling over.

PROVEN, NOT PROMISED

GRADED AGAINST THE OFFICIAL HARNESS

98.2%

official self-test
(1,000 labeled chains)

100%

clean chains —
zero false positives

5/5

attack categories
detected & classified

0

crashes on malformed
or incomplete input

Byte-exact on the organizer's reference crypto. Validated beyond the corpus: **1,500+ freshly-synthesized signed chains** with an independent labeler, plus targeted tests for the gaps the training set never covers — diamond DAGs, anchor-registry replay, designation boundaries.

WHY IT'S DIFFERENT

PROOF WITHOUT EXPOSURE

Provenance you can trust, without leaking the supply chain a competitor could copy. Built on patterns proven in **Catena-X**, the German automotive data space.

Supply chain stays hidden

Verify origin without revealing who the suppliers are or what they charge.

Federated, not centralized

No single honeypot — providers keep their own data and share only what's needed.

Identity + trust anchor

Keyed identities vouched by a trusted authority; a signed anchor registry an attacker can't rewrite.

QR → digital product passport

A buyer-facing scan any non-technical user can read in seconds.

WHY IT MATTERS

SOVEREIGN SUPPLY CHAINS

Drones are the use case — the *model* is the point. AEGIS makes **“Buy Canadian”** enforceable for any multi-tier product.



Protects defence procurement

Gives the Canadian Armed Forces a way to trust origin claims on dual-use goods.



Scales beyond drones

The same attestation graph + stateless verifier works for any layered supply chain — deployed today as a one-command Docker Compose service.



Claim → proof

Turns “Made in Canada” from a label into something a buyer can independently verify in seconds.

OUR TEAM

TAN

Cheick Ismael Maiga

TEAM LEAD

Athif Shaffy

STRATEGY & ARCHITECTURE

Boubacar Ba

LEAD DEVELOPER

Thariq

ENGINEER

Prof. Sk. Md. Mizanur Rahman, PhD., ECSA

ADVISOR

Cryptography, security architecture & data-space guidance — his mentorship shaped the trust model behind AEGIS.



AEGIS

ORIGIN YOU CAN PROVE, NOT JUST CLAIM.

Signed at the source · linked into a tamper-evident chain · independently verifiable.